

# Data Analysis: Predictors of Pokémon HP

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## Response to feedback from Part 1

### Instructor's Comments

In our Part 1 submission we mixed code and output directly into the report, relied heavily on dataset variable names, and presented many tables and plots with minimal narrative context. Based on the feedback we received, we have reorganized Part 2 to read as a stand-alone statistical report rather than an analysis notebook. In particular, we

- move all implementation details and R code out of the written report,
- only includes figures and tables that directly support our hypotheses, and
- describes Pokémon concepts in plain language so that the reader does not need to follow links or infer meaning from raw variable names.

Where Part 1 showed detailed count tables and missing-value summaries, here we summarize those diagnostics in text and focus the main body on interpretation of our model, hypothesis tests, and diagnostics. Overall, Part 2 is structured to emphasize narrative, interpretation, and communication of results at a higher level, following the guidance from the Part 1 feedback.

### Peer-review Comments

Peer reviewers suggested expanding the introduction in Part 1 to provide more context on how Pokémon works, its game mechanics, and why we are interested in HP and its relationship to other parameters. In response, we revised the Part 1 report to include additional background on the game, the meaning of stats such as HP, attack, and speed, and the motivation for focusing on HP as the outcome. Our peer reviewers also recommended being more selective about which tables and figures to include, and providing a brief explanation of what each one shows and why it matters. We incorporated this feedback while preparing this report.

## 1 Model specification

We study how a Pokémon's hit points (HP) depend on other in-game characteristics using a linear regression model. The outcome is HP, and the predictors are:

- base experience;
- attack, defense, special attack, special defense, and speed;
- `dual_type`  $\in \{0, 1\}$  indicating whether the Pokémon has both a primary and secondary type;
- $\log(\text{weight})$  and  $\log(\text{height})$ ;
- the interaction `dual_type`  $\times$  defense.

At the individual level we write the model as

$$\begin{aligned} \text{HP}_i = & \beta_0 + \beta_1 \text{base\_experience}_i + \beta_2 \text{attack}_i + \beta_3 \text{defense}_i \\ & + \beta_4 \text{special\_attack}_i + \beta_5 \text{special\_defense}_i + \beta_6 \text{speed}_i \\ & + \beta_7 \text{dual\_type}_i + \beta_8 \log(\text{weight}_i) + \beta_9 \log(\text{height}_i) \\ & + \beta_{10} \text{dual\_type}_i \times \text{defense}_i + \varepsilon_i, \end{aligned} \quad (1)$$

where  $\varepsilon_i$  are assumed to be independent, mean-zero errors with constant variance. In matrix form, this can be written as

$$\mathbf{y} = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I),$$

and we fit the model via ordinary least squares. The fitted model explains about 94% of the variability in HP (adjusted  $R^2$  is approximately 0.94). Table 1 reports the estimated coefficients, standard errors,  $t$ -statistics, and  $p$ -values. Base experience,  $\log(\text{weight})$ ,  $\log(\text{height})$ , defense, special attack, and speed all have statistically significant effects at the  $\alpha = 0.05$  level, with base experience and the size variables positively associated with HP, and the three defensive/velocity stats negatively associated with HP. Attack, special defense, dual type, and the dual type-by-defense interaction are not individually significant at the 5% level.

Table 1: OLS estimates for the HP regression model.

| Term                       | Estimate | Std. Error | $t$ -statistic | $p$ -value |
|----------------------------|----------|------------|----------------|------------|
| Intercept                  | 61.136   | 3.982      | 15.354         | 1.246e-47  |
| Base experience            | 0.265    | 0.016      | 16.671         | 7.521e-55  |
| Attack                     | 0.018    | 0.027      | 0.684          | 0.494      |
| Defense                    | -0.188   | 0.036      | -5.201         | 2.435e-07  |
| Special attack             | -0.074   | 0.026      | -2.885         | 0.004      |
| Special defense            | -0.048   | 0.032      | -1.507         | 0.132      |
| Speed                      | -0.234   | 0.026      | -9.062         | 7.288e-19  |
| Dual type                  | 4.090    | 3.135      | 1.305          | 0.192      |
| $\log(\text{weight})$      | 1.934    | 0.660      | 2.929          | 0.003      |
| $\log(\text{height})$      | 7.387    | 1.543      | 4.787          | 1.967e-06  |
| Dual type $\times$ defense | -0.049   | 0.040      | -1.230         | 0.219      |

## 2 Hypothesis tests and results

Recall we focus on three main hypotheses:

$$H_0^{(1)} : \beta_{\text{speed}} = \beta_{\text{attack}} = 0, \quad H_0^{(2)} : \beta_{\text{dual\_type} \times \text{defense}} = 0, \quad H_0^{(3)} : \beta_{\text{base\_experience}} = 0,$$

and test these using quadratic-form F-tests constructed from the design matrix  $X$ .

### 2.1 Hypothesis 1: speed vs. attack

We first test whether the coefficients for speed and attack are equal, and then whether they are jointly zero:

$$H_0^{(1a)} : \beta_{\text{speed}} = \beta_{\text{attack}}, \quad H_0^{(1b)} : \beta_{\text{speed}} = 0 = \beta_{\text{attack}}.$$

The resulting F-tests yield the  $p$ -values in Table 2.

Table 2: Joint F-tests for Hypothesis 1.

| Hypothesis                                         | $p$ -value  |
|----------------------------------------------------|-------------|
| $\beta_{\text{speed}} = \beta_{\text{attack}}$     | 5.47e-10    |
| $\beta_{\text{speed}} = 0 = \beta_{\text{attack}}$ | $\approx 0$ |

Both  $p$ -values are essentially zero, so at  $\alpha = 0.05$  we reject both  $H_0^{(1a)}$  and  $H_0^{(1b)}$ . There is overwhelming evidence that the speed and attack coefficients differ, and that at least one of them is nonzero. To see which coefficient is driving this result, we test the two parameters individually:

$$H_0^{(1c)} : \beta_{\text{speed}} = 0, \quad H_0^{(1d)} : \beta_{\text{attack}} = 0.$$

Table 3: Individual F-tests for Hypothesis 1.

| Hypothesis                  | $p$ -value  |
|-----------------------------|-------------|
| $\beta_{\text{speed}} = 0$  | $\approx 0$ |
| $\beta_{\text{attack}} = 0$ | 0.494       |

Table 3 shows that we strongly reject  $H_0^{(1c)}$  but fail to reject  $H_0^{(1d)}$ . Thus, conditional on the other predictors, speed has a clear linear association with HP, whereas attack does not. To further quantify these effects, we construct 95% confidence intervals for the two coefficients:

$$\beta_{\text{speed}} \in [-0.285, -0.184], \quad \beta_{\text{attack}} \in [-0.034, 0.071].$$

The interval for speed lies entirely below zero, in line with the F-test rejecting  $H_0^{(1c)}$ . Increasing speed by 10 points is associated with a decrease of roughly 2–3 HP on average, holding everything else fixed. In contrast, the interval for attack contains 0, indicating no clear evidence that attack affects HP once the other stats are included.

## 2.2 Hypothesis 2: interaction between dual type and defense

Our second hypothesis tests whether the interaction between dual type and defense has any effect on HP:

$$H_0^{(2)} : \beta_{\text{dual\_type} \times \text{defense}} = 0.$$

Table 4: F-test for the dual type  $\times$  defense interaction.

| Hypothesis                                            | <i>p</i> -value |
|-------------------------------------------------------|-----------------|
| $\beta_{\text{dual\_type} \times \text{defense}} = 0$ | 0.219           |

The *p*-value is 0.219, larger than  $\alpha = 0.05$ , so we fail to reject  $H_0^{(2)}$ . A 95% confidence interval for the interaction coefficient is

$$\beta_{\text{dual\_type} \times \text{defense}} \in [-0.127, 0.029],$$

which includes 0 and again suggests that this interaction does not have a statistically detectable effect on HP in this linear specification.

## 2.3 Hypothesis 3: base experience

The final hypothesis concerns base experience:

$$H_0^{(3)} : \beta_{\text{base\_experience}} = 0.$$

Table 5: F-test for the base experience coefficient.

| Hypothesis                            | <i>p</i> -value |
|---------------------------------------|-----------------|
| $\beta_{\text{base\_experience}} = 0$ | $\approx 0$     |

The F-test for  $H_0^{(3)}$  yields a *p*-value essentially equal to zero, providing extremely strong evidence that base experience affects HP. The 95% confidence interval for this coefficient is

$$\beta_{\text{base\_experience}} \in [0.234, 0.296],$$

which lies entirely above zero. An increase of 10 points in base experience is associated with about 2.5–3 additional HP on average, even after adjusting for stats and size.

## 3 Model diagnostics

We assess the adequacy of the linear model using standard residual diagnostics. Figure 1 plots observed HP against fitted HP from the regression. The points lie close to the 45-degree line, with slightly larger deviations only for the highest-HP Pokémon. This agrees with the

high adjusted  $R^2$ , and indicates that the linear model captures the main structure in the data.

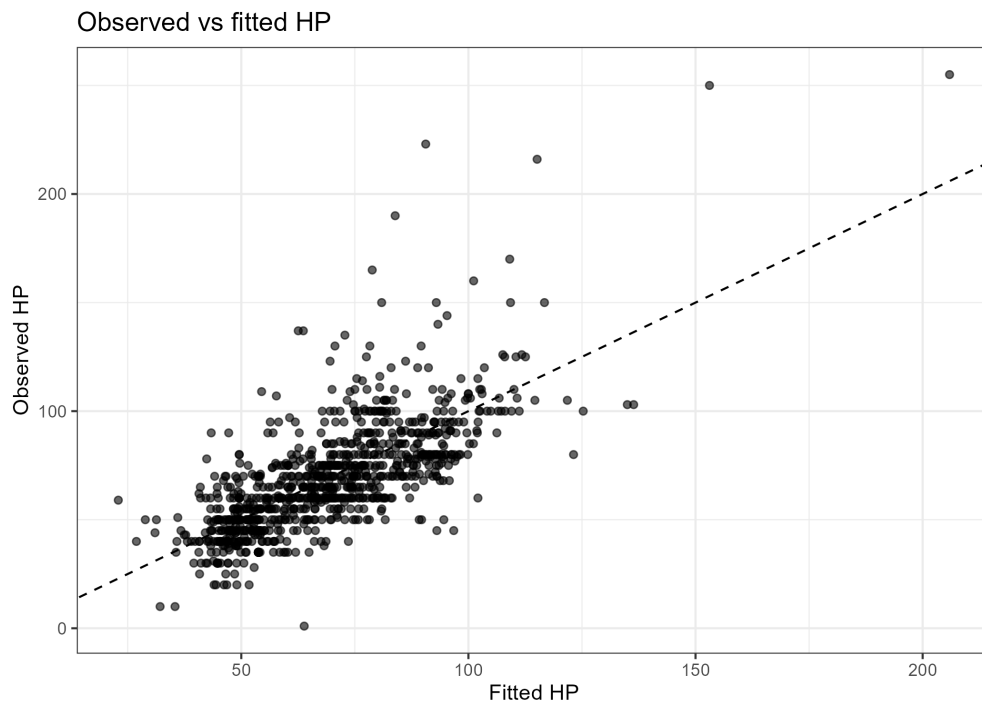


Figure 1: Observed versus fitted HP.

Figure 2 shows residuals versus fitted HP. Residuals are roughly centered around zero across the fitted range, with slightly larger spread for Pokémon with high fitted HP, suggesting mild heteroskedasticity but no strong nonlinearity.

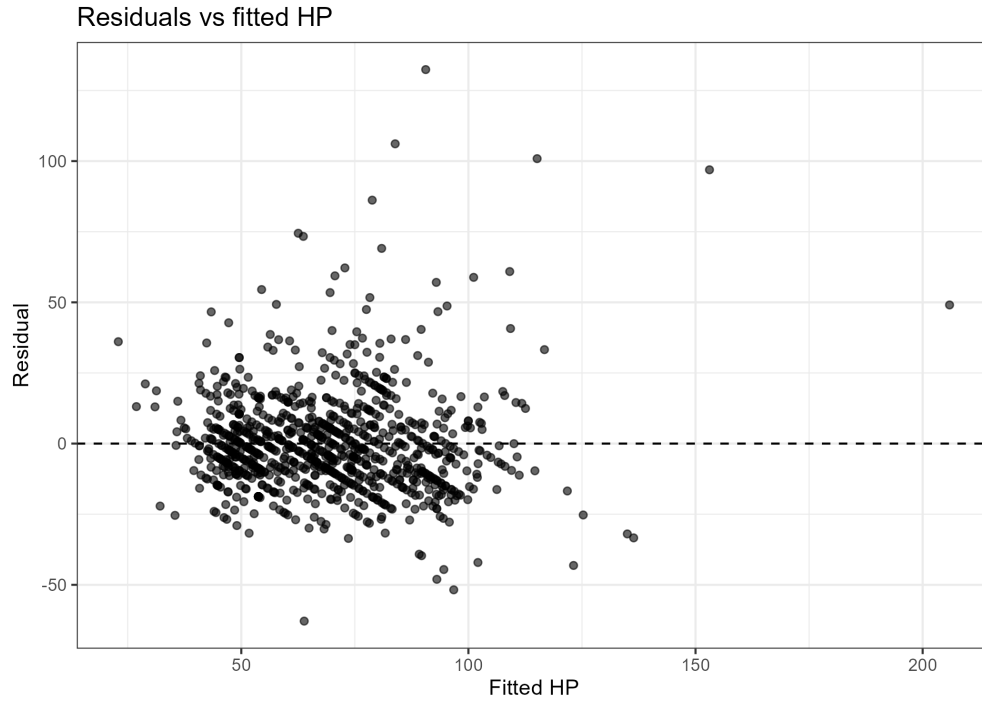


Figure 2: Residuals versus fitted HP.

Figure 3 presents a QQ plot of the residuals. The residual distribution is close to normal in the center, but exhibits heavier upper tails—very large positive residuals occur somewhat more often than under a strict normal model.

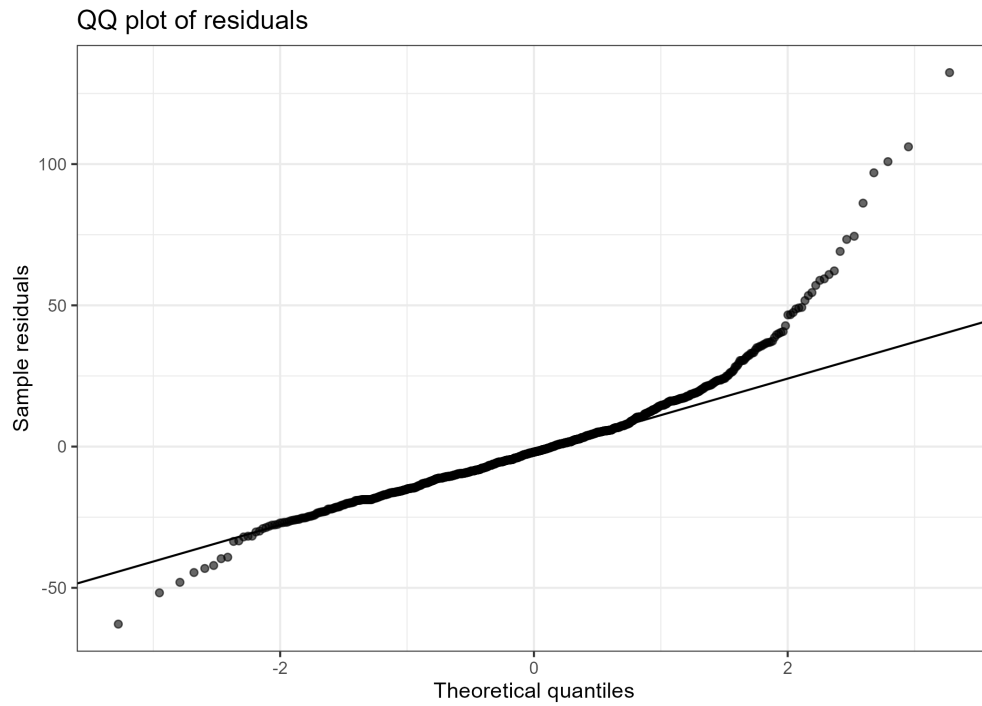


Figure 3: QQ plot of residuals.

Overall, these diagnostics support the use of F inferences in this setting, while cautioning that inferences for the most extreme HP values should be interpreted with care.

## 4 Visual summary of key coefficients

To summarize the key effects visually, Figure 4 displays the point estimates and 95% confidence intervals for the four coefficients that appear in our main hypotheses: speed, attack, the dual type  $\times$  defense interaction, and base experience.

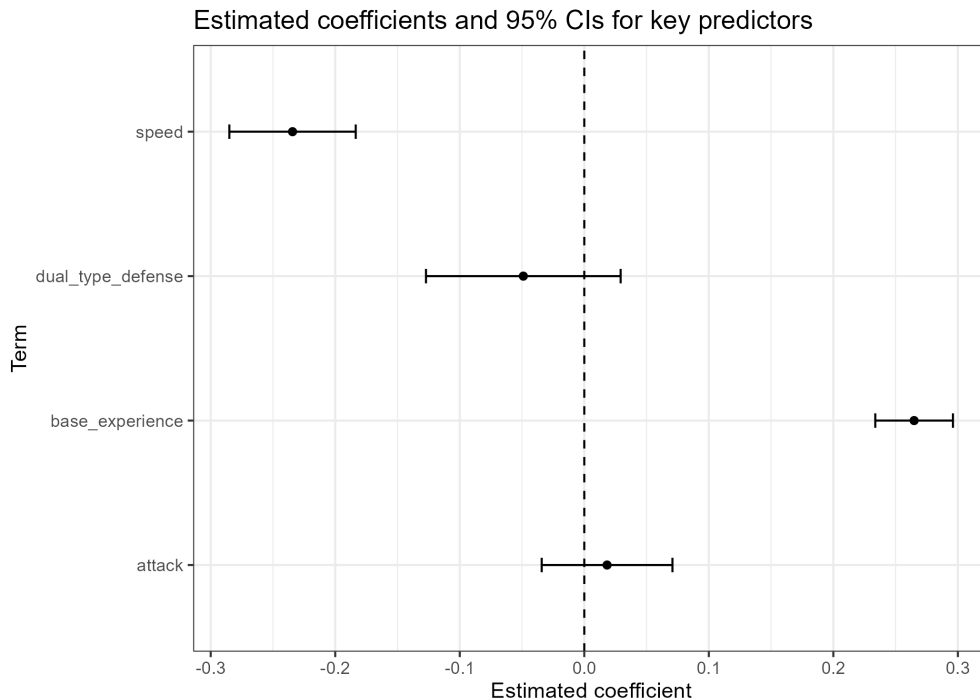


Figure 4: Estimated coefficients and 95% confidence intervals for key predictors.

Two effects stand out clearly. First, the coefficient for speed is negative and precisely estimated:  $\hat{\beta}_{\text{speed}} = -0.234$  with 95% interval  $[-0.285, -0.184]$ . Faster Pokémon tend to have lower HP after adjusting for other characteristics. Second, the coefficient for base experience is positive and precisely estimated:  $\hat{\beta}_{\text{base\_experience}} = 0.265$  with 95% interval  $[0.234, 0.296]$ , confirming that more experienced Pokémon tend to have higher HP.

In contrast, the intervals for attack and the dual type  $\times$  defense interaction both contain zero: for attack we obtain  $[-0.034, 0.071]$ , and for the interaction we obtain  $[-0.127, 0.029]$ . These visual results align with the F-tests: HP is strongly related to speed and base experience after adjustment, but we do not observe clear linear effects of attack or of the dual-type interaction.

## 5 Summary of hypothesis tests

Table 6 collects the p-values and 5% decisions for all hypothesis tests considered above.

Table 6: Summary of hypothesis tests at  $\alpha = 0.05$ .

| Hypothesis                                               | $p$ -value | Decision             |
|----------------------------------------------------------|------------|----------------------|
| H1a: $\beta_{\text{speed}} = \beta_{\text{attack}}$      | 0.000      | Reject $H_0$         |
| H1b: $\beta_{\text{speed}} = 0 = \beta_{\text{attack}}$  | 0.000      | Reject $H_0$         |
| H1c: $\beta_{\text{speed}} = 0$                          | 0.000      | Reject $H_0$         |
| H1d: $\beta_{\text{attack}} = 0$                         | 0.494      | Fail to reject $H_0$ |
| H2: $\beta_{\text{dual.type} \times \text{defense}} = 0$ | 0.219      | Fail to reject $H_0$ |
| H3: $\beta_{\text{base\_experience}} = 0$                | 0.000      | Reject $H_0$         |

The pattern is coherent:

- All tests involving the speed coefficient reject the null, confirming that speed is strongly associated with HP once other variables are controlled for.
- Attack is not significant, and we fail to reject  $H_0^{(1d)}$ , indicating no clear linear effect of attack on HP.
- The interaction between dual type and defense is not significant, suggesting that HP does not depend on defense differently for single-type versus dual-type Pokémon in this model.
- Base experience is highly significant and positively associated with HP; we decisively reject  $H_0^{(3)}$ .

Together with the diagnostics, these results support a simple narrative: HP is most strongly associated with base experience and speed, with additional contributions from defense and special attack, while attack, special defense, dual type, and the dual-type interaction are not statistically important in explaining HP in this linear model.

## 6 Limitations and future extensions

As our peer reviewers noted, one limitation of our analysis is that although this analysis includes t-tests and F-tests, these results should be viewed as descriptive. Any inference on these results should be done with caution because the hypotheses were chosen after examining the data rather than chosen beforehand.

As discussed in Part 1, one limitation of our dataset is that it omits several potentially relevant characteristics, such as catch rate, legendary/mythical status, and evolutionary stage. These quantities may affect both a Pokémon’s usefulness and how its stats are balanced in the game. Including such variables and examining whether they explain additional variation in HP would be a natural extension.



In addition, our analysis is based on a relatively simple linear model. Future work could explore nonlinear relationships (for example, using polynomial terms, interactions beyond dual type and defense, or flexible methods such as generalized additive models) to see whether they capture systematic patterns left in the residuals.

## **Acknowledgements**

ChatGPT was used to assist with code implementation and editing. Yin and Kareena performed the exploratory data analysis and created the figures in Part 1. Kareena implemented the hypothesis test code in Part 2. Yin implemented the diagnostic plots in Part 2. Irene wrote the narrative portions of the reports. Sam prepared the presentation.